A

Major Project

On

A Hybrid AES-ECC Encryption and Image Steganography Framework with MFA for Self -protector Data Security

(Submitted in partial fulfilment of the requirements for the award of Degree)

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In

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CERTIFICATE

This is to certify that the project entitled **"Hybrid AES – ECC Encryption and Image Steganography Framework with Multi-factor Authentication for Self-Protector Data Security"** being submitted by **Hajira Begum (227R1A1221), M. Sathvika (227R1A1242), Asrith (227R1A1207) and Ashfaq (217R1A1296)** in partial fulfilment of the requirements for the award of the degree of B.Tech in Information Technology to the Jawaharlal Nehru Technological University Hyderabad, during the year 2025-26.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree or diploma.

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ABSTRACT

With the rapid expansion of centralized cloud platforms and decentralized storage solutions such as P2P networks and blockchain-based systems, user data frequently resides in remote servers beyond the user's direct control. While most service providers employ strong encryption, sensitive information remains vulnerable to insider threats, particularly malicious administrators who can access stored cryptographic keys and decrypt confidential data. The proposed work introduces a self-protector data security framework that integrates hybrid AES–ECC encryption, Image Steganography, and Multi-Factor Authentication (MFA) to ensure robust, end-to-end protection. In the hybrid encryption process, user files are encrypted using a combination of symmetric Advanced Encryption Standard (AES) for performance efficiency and asymmetric Elliptic Curve Cryptography (ECC) for secure key exchange. This dual-layer approach ensures that no single entity, including server employees, can access both encryption keys, thereby preventing unauthorized decryption. To further conceal sensitive messages, the framework employs image-based steganography, embedding encrypted messages within innocuous images before upload. These steganographic files appear as standard images on the network, making it extremely difficult for malicious actors to detect or extract hidden data. A cryptographic hash code is also generated for each file to enable integrity verification, ensuring that any unauthorized modification is immediately detectable. Access control is enforced through multi-factor authentication, where users first log in with their credentials and then confirm their identity via a one-time password (OTP) sent to their registered email address. This two-step verification prevents account breaches even if login credentials are compromised. Beyond security, the framework incorporates Cybersecurity Awareness Modules, providing users with learning materials and real-time news updates on emerging threats and protective tools, fostering a proactive security mindset. This integrated solution offers confidentiality, integrity, and availability by combining cryptographic robustness with covert data concealment and strict authentication controls. By ensuring that encryption keys, storage mechanisms, and embedded data remain independently secure, the framework significantly mitigates risks of data leakage, tampering, and unauthorized access, making it a comprehensive approach for modern data security in untrusted storage environments.

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1. INTRODUCTION

1. INTRODUCTION

Information security has become increasingly important due to the growing number of cyberattacks targeting businesses, governments, and individuals. As many services shift from local networks to the Internet, protecting the privacy and confidentiality of transmitted data has become essential. To address this challenge, modern security techniques combine encryption and steganography to strengthen data protection. Encryption algorithms increase the complexity of data, making it difficult for attackers to understand even if the data is intercepted.

Steganography enhances security by hiding the existence of secret data inside digital media such as images, audio, or video files. Unlike encryption alone, steganography conceals the presence of the message itself, reducing the chances of detection. The proposed approach combines encryption algorithms like AES and Blowfish with steganography to securely embed confidential messages into images. This method ensures data confidentiality and integrity while maintaining minimal visual difference between the original image and the stego image, making it resistant to visual and analytical attacks.

1.1 Project Purpose

The rapid growth of digital communication and cloud services has increased the risk of sensitive data being exposed to cyber threats. Large volumes of personal, financial, and organizational information are transmitted daily, making traditional single-layer encryption insufficient to handle modern security challenges. To address this issue, hybrid security mechanisms that combine multiple protection techniques are required. Integrating Hybrid Encryption using Elliptic Curve Cryptography (ECC) and Advanced Encryption Standard (AES) with steganography enhances data security by providing secure key exchange, efficient encryption, and concealed data transmission.

The objective of this research is to design a hybrid encryption framework where ECC is used for secure key exchange and AES is used for strong and efficient data encryption. The system ensures data confidentiality and integrity by encrypting sensitive information before storage and transmission. In addition, ECC strengthens key management and protects against unauthorized access, while image steganography is used to hide the encrypted data inside digital images. This

approach enables secure and undetectable communication, making the framework suitable for applications such as cloud storage, digital forensics, and confidential data sharing.

1.2 Project Features

This project enhances data security by integrating hybrid encryption with image steganography to provide a multi-layered protection mechanism. It combines **Elliptic Curve Cryptography (ECC)** for secure key exchange and **Advanced Encryption Standard (AES)** for fast and strong data encryption. In addition, **Steganography** is used to hide the encrypted data within digital images so that the existence of sensitive information remains concealed from unauthorized users.

The system ensures strong confidentiality and integrity by encrypting the data before embedding it into an image and verifying file authenticity using hash verification techniques. An OTP-based authentication mechanism is also implemented to provide secure user access and prevent unauthorized login attempts. This additional authentication layer strengthens the overall security of the platform.

Furthermore, the framework offers efficient and lightweight encryption, improved key management, and secure storage of encrypted files within a structured database. The system minimizes visible changes in the cover image to avoid detection through visual attacks while maintaining high data-hiding capacity. By combining encryption, authentication, and steganography, the proposed approach provides a reliable and scalable solution for protecting sensitive information in environments such as cloud storage, secure communication systems, healthcare data management, and government or organizational records.

2. LITERARTURE SURVEY

2. LITERATURE SURVEY

With the rapid expansion of digital communication systems, cloud computing, and Internet-based services, the need for advanced security mechanisms has become increasingly important. Researchers have explored several approaches that combine cryptographic techniques with steganography to enhance the confidentiality, integrity, and authenticity of sensitive data. Hybrid encryption methods and multi-layer security frameworks have gained considerable attention because they provide stronger protection compared to traditional single-layer security mechanisms.

Pokhrel et al. [6] aimed to address the challenges of maintaining confidentiality in Software Defined Networking (SDN) by introducing a genetic algorithm–based hybrid encryption network policy designed to secure messages transmitted across the network. The proposed technique enhances encryption strength by using optimization techniques from genetic algorithms to generate more complex encryption patterns. Experimental results indicated that the proposed hybrid mechanism achieved an average entropy value of 0.989, which demonstrates a high level of randomness and improved resistance against cryptanalysis attacks. Despite these advantages, the approach introduced noticeable processing overhead, which significantly increased the transmission time for encrypted messages when compared to unencrypted communication. While the proposed model performed better than standalone encryption algorithms such as AES, DES, and RSA in terms of encryption randomness, the study also highlighted the trade-off between security strength and network performance due to the absence of load-balancing strategies.

Alanzy et al. [7] proposed a Multi-Level Steganography (MLS) algorithm that integrates two powerful encryption algorithms, AES and Blowfish, to enhance the protection of confidential information embedded in digital images. In their approach, encryption keys are stored as key images within the stego image itself, which adds an additional security layer. The algorithm incorporates a pixel randomization function that distributes secret data across the image in a non-sequential manner, thereby increasing the complexity of data extraction by attackers. Experimental evaluation demonstrated that the proposed MLS technique produces high-quality stego images with high Peak Signal-to-Noise Ratio (PSNR) and low Mean Square Error (MSE) values. These metrics confirm that the visual quality of the image remains almost unchanged even after data

embedding. The integration of AES and Blowfish further strengthens the encryption complexity, ensuring that the hidden message remains secure even if the stego image is intercepted.

Adee et al. [8] focused on developing a data security model that integrates cryptography and steganography for protecting data stored in cloud computing environments. Cloud storage systems are particularly vulnerable to threats such as data loss, unauthorized access, data manipulation, and data theft. To address these issues, the researchers applied a design science research methodology to analyze existing security models and identify the core challenges associated with cloud security. The methodology included several stages such as problem identification, requirement analysis, artifact design and development, system demonstration, and performance evaluation. The proposed solution was implemented using the Python programming language and evaluated through graphical representations such as histograms and performance tables. The study demonstrated that combining encryption with steganographic data hiding can significantly improve the protection of sensitive information stored in cloud systems.

Kallapu et al. [9] proposed a comprehensive multi-layered security framework that integrates various cryptographic and steganographic techniques to strengthen data protection. The proposed framework includes methods such as encrypted data embedding in images, DNA sequence coding, QR code generation, and Least Significant Bit (LSB) steganography. These techniques collectively provide multiple layers of data protection and enhance the difficulty of unauthorized data extraction. The effectiveness of the framework was evaluated using various types of input data, including text messages, text files, and images. Several security performance metrics were analyzed, such as PSNR, MSE, Signal-to-Noise Ratio (SNR), and encryption–decryption processing time. In addition, image security was evaluated through visual inspection, correlation analysis, entropy measurement, key space analysis, randomness testing, and differential attack analysis. The experimental results showed strong resistance against differential cryptanalysis attacks, with high NPCR values of 99.5784%, 99.4292%, and 99.5784%, and UACI values of 33.5873%, 33.5149%, and 33.3745%. These results indicate strong diffusion and confusion properties, which are essential characteristics of secure encryption systems.

Infantia Henry et al. [10] presented a secure framework for exchanging data through cloud platforms while maintaining both the confidentiality and integrity of transmitted information. Their approach combines Elliptic Curve Cryptography (ECC) with the Advanced Encryption

Standard (AES) to create a hybrid encryption mechanism. ECC is utilized for secure key generation and exchange, while AES performs high-speed symmetric encryption of the actual data. This combination ensures that encrypted data remains protected even if communication channels are compromised. The study demonstrates that hybrid encryption approaches can significantly enhance data protection in distributed cloud environments.

Similarly, Chowdhary et al. [11] proposed a hybrid encryption model that integrates ECC and AES to improve the reliability of secure data communication. The proposed method focuses on verifying data authenticity and preserving the integrity of sensitive information transmitted through cloud-based platforms. The results demonstrate that combining asymmetric encryption with symmetric encryption can improve both security performance and computational efficiency, making it suitable for large-scale cloud applications.

Chen et al. [12] conducted a comparative analysis of different hybrid encryption techniques for secure image encryption and decryption. Their study evaluated combinations such as ECC with Hill Cipher, ECC with AES, and ElGamal with Double Playfair Cipher. The performance of these encryption combinations was analyzed using multiple evaluation parameters including encryption time, decryption time, entropy of encrypted images, Peak Signal-to-Noise Ratio (PSNR), Number of Pixel Change Rate (NPCR), and Unified Average Changing Intensity (UACI). The results demonstrated that hybrid encryption approaches provide better resistance against statistical and differential attacks compared to traditional encryption algorithms.

Overall, the reviewed literature highlights the growing importance of hybrid security frameworks that combine encryption, steganography, artificial intelligence, and authentication mechanisms. These studies demonstrate that multi-layered security approaches significantly improve data confidentiality, integrity, and resistance against modern cyber threats, forming the foundation for developing more secure and efficient data protection systems.

2.1 REVIEW OF RELATED WORK

The rapid development of digital communication, cloud computing, and data sharing technologies has significantly increased the risk of cyberattacks and unauthorized access to sensitive information. As a result, researchers have focused on developing advanced security mechanisms to ensure the confidentiality, integrity, and authenticity of data. Various approaches

such as cryptographic algorithms, steganography techniques, and hybrid security frameworks have been proposed to strengthen data protection. This section reviews existing research related to hybrid encryption and steganography-based security systems.

2.1.1 Hybrid Encryption-Based Security Approaches

Several studies have explored hybrid encryption techniques to improve the security of digital communication systems. Hybrid encryption combines the advantages of symmetric and asymmetric encryption methods. Asymmetric encryption algorithms are used for secure key exchange, while symmetric encryption algorithms are used for efficient data encryption.

Pokhrel et al. proposed a hybrid encryption approach based on genetic algorithms to enhance message confidentiality in Software Defined Networking (SDN). The proposed model improved encryption randomness and achieved a high entropy value, indicating strong encryption performance. However, the study also reported increased computational overhead and longer transmission times due to the complex encryption process.

Other researchers have proposed hybrid models combining Elliptic Curve Cryptography (ECC) and Advanced Encryption Standard (AES). In these approaches, ECC is used to securely generate and exchange encryption keys, while AES is used to encrypt large volumes of data efficiently. These hybrid encryption models provide stronger protection against cryptographic attacks while maintaining better computational performance compared to traditional encryption methods.

2.1.2 Steganography-Based Data Hiding Techniques

Steganography is a widely used technique for hiding secret data within digital media such as images, audio files, and videos. Unlike encryption, which transforms data into an unreadable format, steganography hides the very existence of the data by embedding it within a cover medium. This makes it difficult for attackers to detect the presence of sensitive information.

Researchers have developed several image steganography techniques, including Least Significant Bit (LSB) embedding, pixel value modification, and pixel randomization methods. These techniques aim to hide secret data within image pixels while maintaining the visual quality

of the image. Performance of steganographic methods is typically evaluated using parameters such as Peak Signal-to-Noise Ratio (PSNR), Mean Square Error (MSE), and Structural Similarity Index (SSIM). High PSNR and low MSE values indicate minimal distortion in the stego image.

Some advanced steganography models also incorporate multi-level data hiding techniques, where encrypted data is embedded within images using multiple layers of security. These approaches significantly increase the difficulty for attackers attempting to detect or extract hidden information.

2.1.3 Multi-Layer Security Frameworks

To further enhance data protection, several studies have proposed multi-layer security frameworks that integrate both encryption and steganography techniques. In these systems, sensitive data is first encrypted using strong cryptographic algorithms and then embedded into digital images or other media using steganographic techniques.

For instance, some researchers proposed frameworks that combine encryption algorithms with techniques such as DNA coding, QR code encoding, and image steganography. These systems create multiple layers of security, making unauthorized access extremely difficult. Security performance of these systems is evaluated using various metrics such as entropy analysis, correlation analysis, key space analysis, and resistance to differential attacks.

2.1.4 Emerging Technologies in Data Security

Recent research has also explored the use of advanced technologies such as artificial intelligence, deep learning, and bio-inspired encryption techniques to enhance data security. Some researchers have proposed steganography systems based on deep neural networks that can automatically learn effective data embedding patterns. Others have explored biological structures such as DNA sequences and protein chains for encryption and data hiding.

Additionally, modern privacy-preserving models have been proposed for Internet of Things (IoT) and Internet of Medical Things (IoMT) environments to ensure secure communication between connected devices. These systems aim to provide lightweight encryption and efficient data protection while maintaining low computational overhead.

From the review of existing literature, it can be observed that hybrid encryption and steganography techniques play a crucial role in enhancing data security. While many existing approaches provide strong protection against cyber threats, challenges such as computational complexity, scalability, and efficient key management still remain. Therefore, developing an efficient hybrid framework that integrates encryption and steganography while maintaining high security and performance remains an important research direction.

2.2 DEFINITION OF PROBLEM STATEMENT

In the modern digital era, the widespread use of cloud computing, online communication platforms, and digital storage systems has led to an enormous increase in the exchange of sensitive information. Organizations and individuals regularly transmit confidential data such as personal records, financial transactions, medical information, and business documents through the internet. However, this rapid growth in digital data exchange has also increased the risk of cyberattacks, data breaches, and unauthorized access to confidential information.

Traditional security methods mainly rely on single encryption techniques to protect sensitive data during storage and transmission. Although encryption algorithms can convert data into unreadable formats, encrypted data can still attract attackers' attention, making it a potential target for cryptanalysis attacks. In many cases, once encrypted data is intercepted, attackers attempt to break the encryption using advanced computational methods. This creates a significant challenge in maintaining long-term data confidentiality and protection.

Another major issue is the lack of multi-layer security mechanisms in many existing data protection systems. Many systems fail to provide secure key exchange, strong authentication, and protection against data tampering. Furthermore, sensitive information transmitted through cloud platforms and communication networks may be exposed to interception, modification, or leakage if proper security measures are not implemented.

To overcome these challenges, it is necessary to design a robust and efficient data protection mechanism that not only encrypts sensitive data but also conceals its presence during transmission. Integrating hybrid encryption techniques with steganography can provide multiple layers of security, ensuring that confidential information remains protected from unauthorized

access, interception, and modification. Such a system can significantly enhance data confidentiality, integrity, and secure communication in modern digital environments.

2.3 EXISTING SYSTEM

The existing system for managing and securing information is primarily based on a manual, non-computerized approach in which all operations depend on human effort and physical documentation. In this system, data collection is carried out using handwritten or printed paper forms that are submitted by users. Identity verification is performed manually through physical identification documents, signatures, or direct verification by authorized personnel. Authorization for accessing or modifying records depends on hierarchical approval from designated staff members, which often slows down the overall process.

Once verification is completed, the collected information is manually recorded in registers or stored in paper files. This manual transcription process may introduce delays, human errors, and inconsistencies in data recording. Records are typically organized and classified according to categories, dates, or departments, and are stored physically in cabinets, shelves, or record rooms. These storage locations are secured using conventional locking mechanisms such as locks and keys. Access to records is limited to specific locations and requires the physical presence of authorized personnel.

Searching for particular information in a manual system requires staff members to inspect files and registers individually, which makes the retrieval process time-consuming and unpredictable. Updating records is also difficult, as it may involve overwriting existing entries or creating duplicate records, which can result in data inconsistency and confusion. The manual system does not provide technical mechanisms such as encryption, digital authentication, or automated security controls. Instead, confidentiality is maintained through administrative rules and procedural policies rather than technological safeguards.

In addition, the scalability of the manual system is limited because increasing volumes of data require more physical storage space and additional manpower for management. Data sharing between departments typically requires the physical transfer of documents, which can further delay operations. The system does not support real-time data updates or remote access, and overall operational efficiency decreases significantly as the amount of stored information increases.

LIMITATIONS OF EXISTING SYSTEM

- **Lack of Technical Security Mechanisms:** The existing system does not include technical security features such as encryption, authentication, or access control. As a result, sensitive information is not digitally protected and can be easily accessed or exposed if proper physical safeguards are not maintained.
- **High Dependency on Human Integrity:** The system relies heavily on human honesty and discipline to maintain the confidentiality and safety of information. Since there are no automated controls or monitoring mechanisms, the effectiveness of the system depends entirely on the responsible behavior of staff members.
- **Risk of Data Tampering:** Manual records can be easily modified, altered, or manipulated without leaving a trace. The absence of automated tracking or logging makes it difficult to detect unauthorized changes to the data.
- **Vulnerability to Physical Damage or Loss:** Since data is stored in physical form, it is highly vulnerable to damage or loss caused by fire, theft, moisture, natural disasters, or mishandling. Such incidents may result in the permanent loss of valuable information.
- **Absence of Real-Time Monitoring and Auditing**: The manual system does not support real-time monitoring of data access or modifications. Without automated auditing mechanisms, it becomes difficult to track who accessed the data and when the access occurred.

2.4 PROPOSED SYSTEM

The proposed system provides a secure framework for protecting sensitive data using a combination of hybrid encryption, image steganography, and multi-factor authentication. The system is designed to overcome the limitations of traditional single-layer security methods by introducing multiple layers of protection to ensure confidentiality, integrity, and secure access to data.

In this system, users first register and log in through a secure authentication process that includes OTP-based Multi-Factor Authentication (MFA). After successful authentication, the user can upload sensitive files or messages. The uploaded data is encrypted using the Advanced Encryption Standard (AES), which converts the original data into unreadable ciphertext. To

enhance security, the AES encryption key is further protected using Elliptic Curve Cryptography (ECC), ensuring secure key management and preventing unauthorized access.

When an authorized user requests the stored data, the system extracts the hidden encrypted content from the image, decrypts the AES key using ECC, and finally restores the original data through AES decryption. This layered security approach ensures safe storage, secure transmission, and controlled access to confidential information.

ADVANTAGES OF PROPOSED SYSTEM

- **Enhanced Data Security:** The proposed system provides strong protection for sensitive information by using multiple security layers such as encryption, steganography, and authentication. This significantly reduces the chances of unauthorized access or cyberattacks.
- **Hybrid Encryption Mechanism**: The system uses a combination of AES and ECC algorithms. AES ensures fast and efficient encryption of data, while ECC secures the encryption keys through asymmetric cryptography, providing stronger overall protection.
- **Concealment of Encrypted Data**: By using image steganography with the Least Significant Bit (LSB) technique, the encrypted data is hidden within digital images. This makes the existence of the confidential information difficult to detect by attackers.
- **Multi-Factor Authentication (MFA):** The system includes OTP-based authentication during login. This additional verification step ensures that only authorized users can access or perform operations on the data.
- **Secure Data Storage and Retrieval:** The encrypted files and related metadata are stored securely in the database and file system. Only authenticated users can retrieve and decrypt the stored data, ensuring controlled and secure access.

2.5 OBJECTIVES

The main objective of this project is to design and develop a secure data protection framework that safeguards sensitive information during storage and transmission.

- **To Enhance Data Security**: The primary objective of the project is to improve the overall security of sensitive data by implementing multiple layers of protection, reducing the possibility of unauthorized access and cyber threats.

- **To Implement a Hybrid Encryption Mechanism**: The project aims to develop a hybrid encryption framework that combines AES and ECC algorithms. AES provides fast and efficient data encryption, while ECC ensures secure key exchange and key management.
- **To Conceal Encrypted Data Using Steganography**: Another objective is to hide encrypted data within digital images using the Least Significant Bit (LSB) steganography technique, making the presence of confidential information difficult to detect.
- **To Strengthen User Authentication**: The system focuses on enhancing access control by implementing Multi-Factor Authentication (MFA) through OTP verification, ensuring that only authorized users can access the system.
- **To Provide Secure Data Storage and Retrieval**: The project also aims to ensure that encrypted files and related metadata are securely stored and can only be retrieved and decrypted by authenticated users, maintaining data confidentiality and integrity.

2.6 SYSTEM REQUIREMENTS

2.6.1 Hardware Requirements

Hardware interfaces specify the logical characteristics of each interface between the software product and the hardware components of the system. The following are some hardware requirements.

- Processor : Intel i5
- Memory : 512GB (storage)
- RAM : 8 GB

2.6.2 Software Requirements

Software Requirements specifies the logical characteristics of each interface and software components of the system. The following are some software requirements.

- Python 3.7.6
- Operating system : Windows
- Frontend : Tkinter
- Database : MySQL

3. SYSTEM ARCHITECTURE & DESIGN

3. SYSTEM ARCHITECTURE AND DESIGN

The system architecture integrates authentication, hybrid encryption, and steganography to provide secure data protection. The application is developed using Django with MySQL as the backend database. Data is encrypted using Advanced Encryption Standard (AES), the key is secured with Elliptic Curve Cryptography (ECC), and the encrypted data is hidden inside images using steganography for additional security.

3.1 PROJECT ARCHITECTURE

The architecture defines the overall structure and workflow of the system, showing how different components interact to perform secure data operations. It integrates authentication, encryption, steganography, and database management to ensure secure storage and transmission of sensitive information.

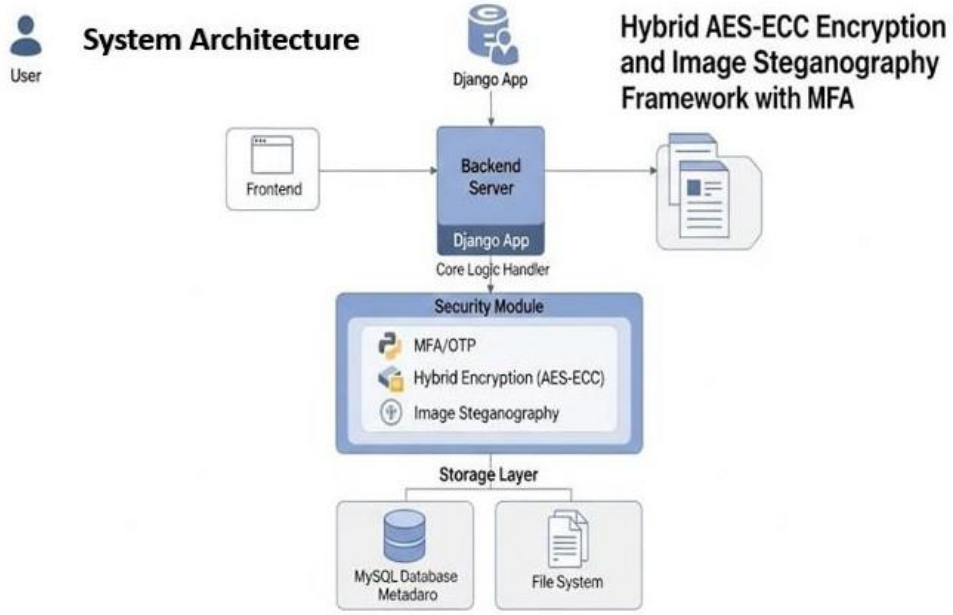

FIG 3.1: Project Architecture

3.2 DESCRIPTION

Input Data: The system receives sensitive files or messages from the user through the frontend interface. The user must first log in and verify their identity using OTP-based authentication before uploading any data. Once verified, the uploaded file is sent to the backend server for further secure processing.

Authentication Module: This module verifies the identity of users through login credentials and Multi-Factor Authentication (OTP verification). It ensures that only authorized users can access the system and perform operations such as file upload, encryption, and data retrieval.

Hybrid Encryption Module: After authentication, the uploaded data is encrypted using a hybrid encryption approach. The data is first encrypted using Advanced Encryption Standard (AES) for fast symmetric encryption, and the AES key is then secured using Elliptic Curve Cryptography (ECC) to ensure safe key exchange and protection.

Steganography Module: Once encryption is completed, the encrypted data is hidden inside an image using the Least Significant Bit (LSB) image steganography technique. This process conceals the presence of the encrypted information by embedding it within image pixels without affecting the visible appearance of the image.

Storage Module: The processed files are stored securely in the system. File metadata such as file name, hash code, and upload details are stored in the MySQL database, while the encrypted or stego-image files are saved in the server file system.

Data Retrieval Module: When an authorized user requests the stored file, the system retrieves it from storage. If steganography was used, the hidden data is first extracted from the image, then the AES key is decrypted using ECC, and finally the original data is recovered through AES decryption. This ensures secure and accurate data access.

3.3 UNIFIED MODELING LANGUAGE(UML)

Unified Modeling Language (UML) is not just a static set of diagrams; it's a versatile toolset that supports multiple views and perspectives of a system. This versatility allows UML to be applied at different stages of the software development lifecycle, from initial requirements

analysis and design to implementation, testing, and maintenance. UML diagrams serve as a common visual language that bridges the gap between technical and non-technical stakeholders. They provide a means for software engineers to communicate complex system designs in a clear and concise manner, facilitating better understanding and collaboration among team members.

GOALS OF UML

1. Provide users a ready-to-use, expressive visual modeling language so that they can develop and exchange meaningful models.

2. Provide extendibility and specialization mechanisms to extend the core concepts.

3. Be independent of particular programming languages and development processes.

4. Provide a formal basis for understanding the modeling language.

3.3.1 USE CASE DIAGRAM

The proposed system's use case diagram, however, includes additional steps like OTP verification, AES encryption, ECC key encryption, and steganography, showing the multi-layered security approach. Each use case represents a discrete function the system provides, highlighting how users interact with both simple and advanced operations**.**

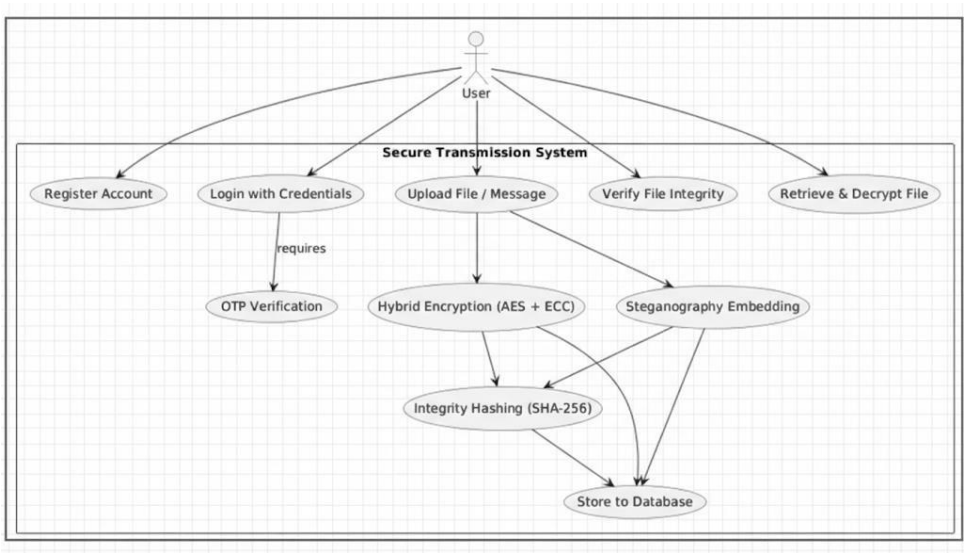

FIG 3.2 : Use Case Diagram

3.3.2 CLASS DIAGRAM

This class diagram represents core classes like User, Authentication, Hybrid Encryption, Steganography, Integrity, and File Storage. Each class contains specific methods that define its role. The relationships show that users interact with different modules to perform secure operations like encryption, message hiding, and file handling.

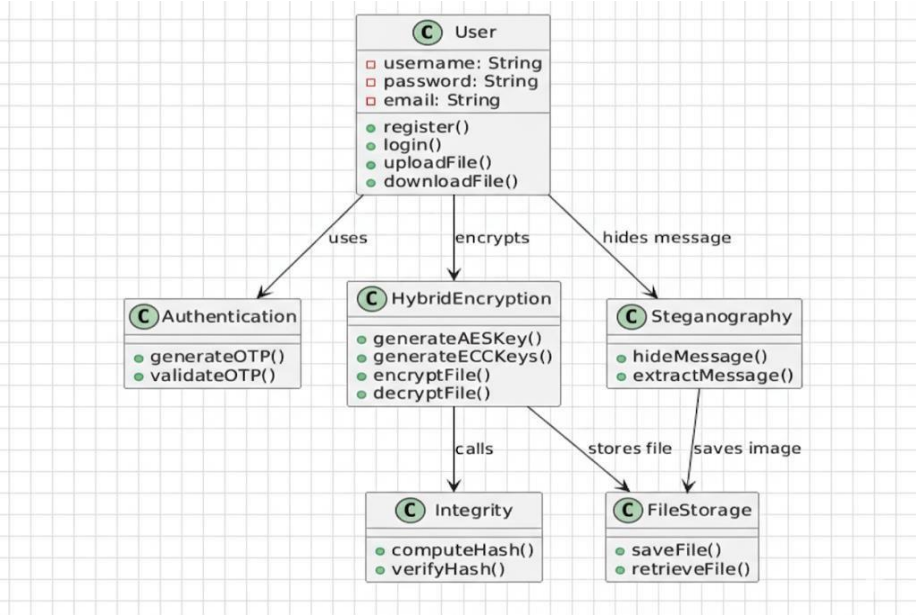

FIG 3.3: Class Diagram

3.3.3 SEQUENCE DIAGRAM

The proposed system sequence diagram depicts a more complex flow, including AES encryption, ECC key encryption, optional steganography, and secure file storage. This diagram helps in understanding how different components collaborate during a user request and the order in which security processes are executed. It also demonstrates how the proposed system handles sensitive data securely throughout the lifecycle. By analysing the sequence diagram, developers can identify potential bottlenecks or points where encryption and decryption occur. Overall, it provides a detailed view of the internal process flow for both the traditional and hybrid systems.

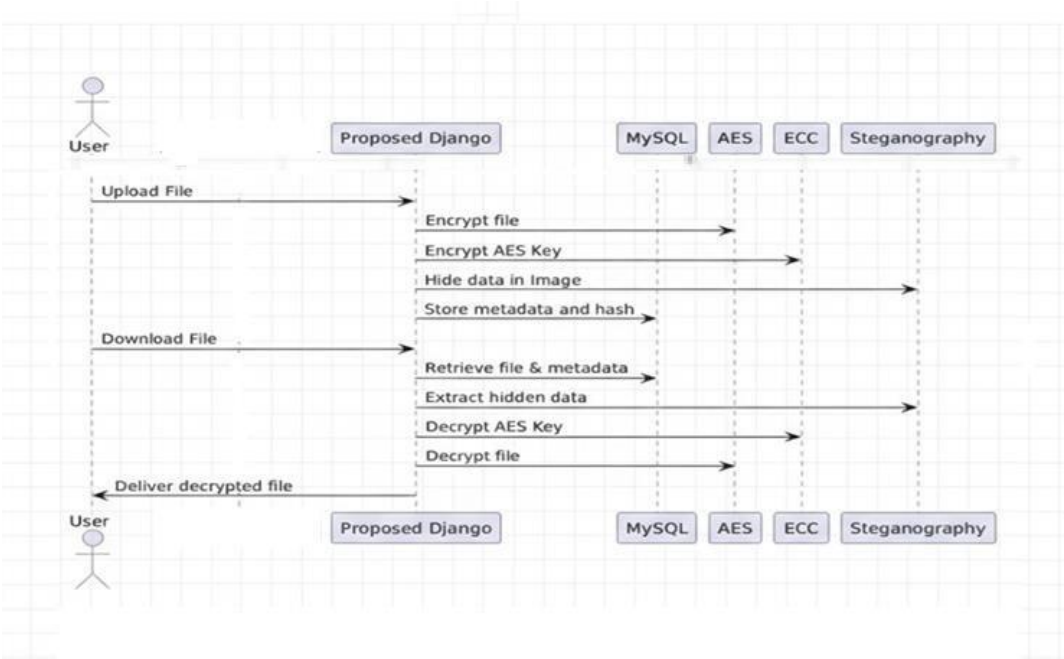

FIG 3.4: Sequence Diagram

3.3.4 ACTIVITY DIAGRAM

The activity diagram represents the workflow of processes within the system. For the existing system, it shows a linear sequence of user login, file upload, storage in MySQL, and file retrieval.

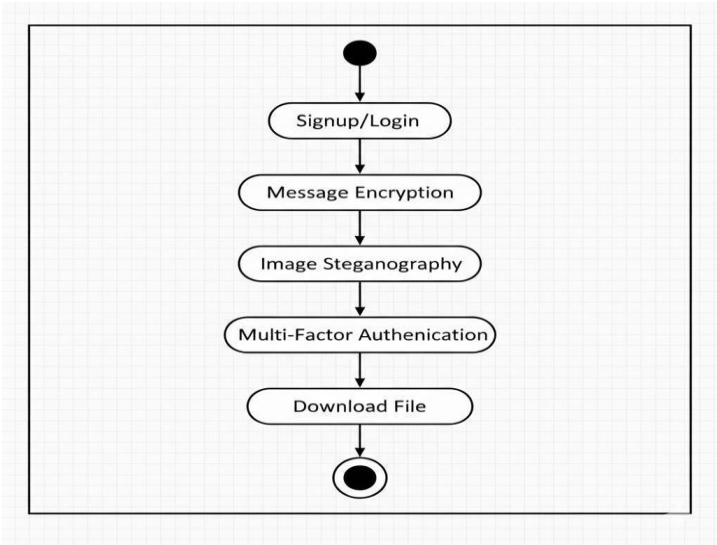

FIG 3.5: Activity Diagram

3.3.5 DATA FLOW DIAGRAM

The following diagram illustrates the overall flow of information within the system and explains how different components interact to process user requests. It shows how data moves between the user, web application, security module, and database during operations such as uploading, storing, and retrieving files. The flow highlights the differences between the existing approach and the proposed secure framework. By introducing encryption, steganography, and additional security processing, the proposed system ensures that sensitive data is protected during both storage and transmission.

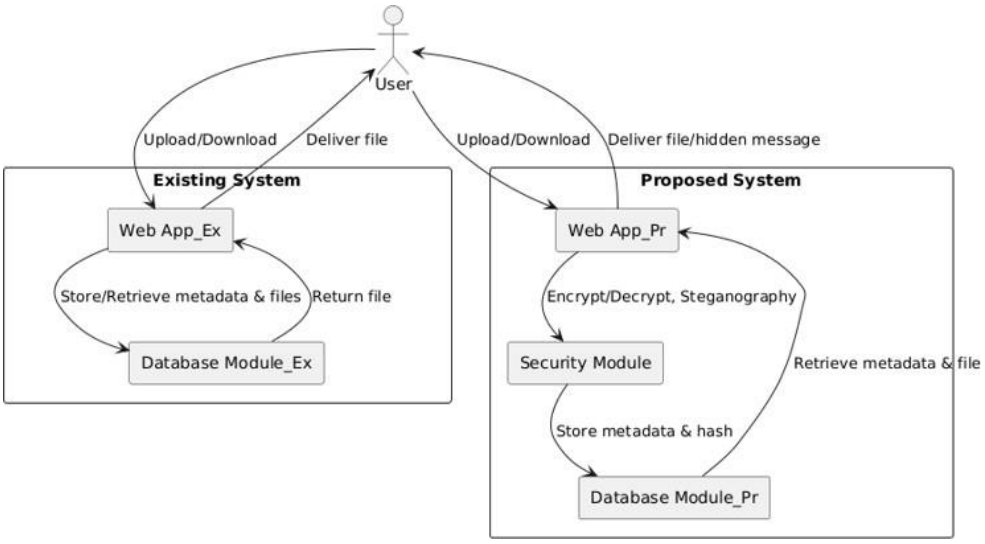

FIG 3.6: Data Flow Diagram

The diagram illustrates the flow of information between the user, web application, security module, and database in both the existing and proposed approaches. It highlights how data is processed during file upload and download operations. In the existing approach, the web application directly interacts with the database to store and retrieve files and metadata. In contrast, the proposed approach introduces an additional security module that performs encryption and steganography before storing data, thereby improving data confidentiality and protection.

User Interaction: The process begins when the user uploads or downloads a file through the web application interface. The user sends requests such as file upload, download, or retrieval of hidden messages.

Proposed System Web Application: In the proposed approach, the user request is first handled by the web application. The application manages file upload and download requests and forwards the data to the security module for further protection.

Security Module Processing: The security module performs encryption, decryption, and steganography operations to protect sensitive data. The file is encrypted using Advanced Encryption Standard (AES), and the encryption key is secured using Elliptic Curve Cryptography (ECC). The encrypted data may also be embedded into an image using steganography for additional concealment.

Database Storage: After security processing, the system stores metadata and hash values in the MySQL database. The encrypted or steganographic files are stored securely in the system.

File Retrieval: When the user requests a file, the system retrieves the stored metadata and encrypted file from the database. The security module then performs decryption and extraction processes before delivering the original file or hidden message to the user. This ensures secure and controlled access to the stored data.

4. IMPLEMENTATION

4. IMPLEMENTATION

The implementation phase of a project involves executing the planned strategies and tasks. It requires meticulous coordination, resource allocation, and monitoring to ensure that objectives are met efficiently. Effective implementation is crucial for achieving project goals and delivering expected outcomes within the set timeline and budget constraints

4.1 ALGORITHMS USED

The proposed secure data protection framework utilizes multiple cryptographic and data hiding algorithms to ensure confidentiality, integrity, and secure communication. These algorithms work together to protect sensitive information during storage and transmission.

1. Advanced Encryption Standard (AES):

Advanced Encryption Standard (AES) is a symmetric encryption algorithm widely used for securing sensitive digital information. In the proposed system, AES is used to encrypt user data before it is stored or transmitted through the network. The algorithm converts readable plaintext into encrypted ciphertext using a secret key, ensuring that only authorized users possessing the correct key can access the original information. AES operates through multiple rounds of substitution and permutation processes that significantly strengthen the security of the data. Due to its high efficiency and strong cryptographic design, AES is considered one of the most reliable encryption standards used in modern data security applications

2. Elliptic Curve Cryptography (ECC):

Elliptic Curve Cryptography (ECC) is an asymmetric encryption technique used for secure key generation and key exchange. In the proposed system, ECC is used to securely encrypt and manage the encryption keys used by the AES algorithm. ECC operates using a pair of keys: a public key for encryption and a private key for decryption. This ensures that only authorized users with the appropriate private key can access the encrypted data. ECC provides a high level of security while requiring smaller key sizes compared to traditional public-key cryptography methods, making it efficient and suitable for modern secure communication systems.

3. Image Steganography (LSB Technique):

Image steganography is a technique used to hide confidential data within digital images. In the proposed system, the encrypted data generated by the AES algorithm is embedded into an image using the Least Significant Bit (LSB) technique. This method modifies the least significant bits of image pixels to store hidden information without significantly altering the visual appearance of the image. As a result, the image appears normal while secretly containing encrypted data. This technique provides an additional security layer by concealing the existence of sensitive information during transmission.

4. Hashing Algorithm (SHA):

Hashing algorithms such as Secure Hash Algorithm (SHA) are used to generate a fixed-length hash value from input data. In the proposed system, hashing is used to generate secure hash values for metadata and files to ensure data integrity. When a file is stored or transmitted, a hash value is generated and stored along with it. During retrieval, the system recomputes the hash value and compares it with the stored hash to verify whether the data has been modified. This mechanism helps detect any unauthorized changes to the data.

5. Multi-Factor Authentication (MFA):

Multi-Factor Authentication (MFA) is a security mechanism that requires users to verify their identity using multiple authentication factors before accessing the system. In the proposed framework, MFA strengthens the authentication process by combining different verification methods such as passwords, one-time passwords (OTP), or other authentication factors. This ensures that even if one authentication factor is compromised, unauthorized users cannot gain access to the system. MFA significantly improves the security of user accounts and prevents unauthorized access to sensitive data.

4.2 SAMPLE CODE

```
from django.shortcuts import render

from django.template import RequestContext

from django.contrib import messages
```

```
from django.http import HttpResponse

from django.core.files.storage import FileSystemStorage

import os

import random

from datetime import date

import ecdsa

from hashlib import sha256

import pickle

import re

import pyaes, pbkdf2, binascii, os, secrets

import pymysql

import smtplib

import hashlib

from ecies.utils import generate_eth_key, generate_key

from ecies import encrypt, decrypt

from PIL import Image


global username, otp, email


#function to get ECC assymetric keys

def getECCKeys():

    if os.path.exists("SecurityApp/static/keys/pvt.key"):

        with open("SecurityApp/static/keys/pvt.key", 'rb') as f:
```

```
            private_key = f.read()
        f.close()
        with open("SecurityApp/static/keys/pri.key", 'rb') as f:
            public_key = f.read()
        f.close()
        private_key = private_key.decode()
        public_key = public_key.decode()
    else:
        secret_key = generate_eth_key()
        private_key = secret_key.to_hex()  # hex string
        public_key = secret_key.public_key.to_hex()
        with open("SecurityApp/static/keys/pvt.key", 'wb') as f:
            f.write(private_key.encode())
        f.close()
        with open("SecurityApp/static/keys/pri.key", 'wb') as f:
            f.write(public_key.encode())
        f.close()
    return private_key, public_key


#function to get AES symmetric keys
def generateKeys():
    if os.path.exists("SecurityApp/static/keys/key.pckl"):
        f = open("SecurityApp/static/keys/key.pckl", 'rb')
```

```
        keys = pickle.load(f)
        f.close()
        secret_key = keys[0]
        private_key = keys[1]
    else:
        secret_key = ecdsa.SigningKey.generate(curve=ecdsa.SECP256k1, hashfunc=sha256) # The
default is sha1
        private_key = secret_key.get_verifying_key()
        keys = [secret_key, private_key]
        f = open("SecurityApp/static/keys/key.pckl", 'wb')
        pickle.dump(keys, f)
        f.close()
    private_key = private_key.to_string()[0:32]
    return private_key


def encryptAES(plaintext, key): #AES data encryption
    aes=pyaes.AESModeOfOperationCTR(key,
pyaes.Counter(31129547035000047302952433967654195398124239844566322884172163637
846056248223))
    ciphertext = aes.encrypt(plaintext)
    return ciphertext


def decryptAES(enc, key): #AES data decryption
```

```
    aes = pyaes.AESModeOfOperationCTR(key,
pyaes.Counter(31129547035000047302952433967654195398124239844566322884172163637
846056248223))
    decrypted = aes.decrypt(enc)
    return decrypted

#ECC based data encryption
def ECCEncrypt(plainText, public_key):
    ecc_encrypt = encrypt(public_key, plainText)
    return ecc_encrypt

#ECC based data decryption
def ECCDecrypt(encrypt, private_key):
    ecc_decrypt = decrypt(private_key, encrypt)
    return ecc_decrypt

def generateBits(message):
    msg_bit = []
    for bit in message:
        binary_bit = format(ord(bit), '08b')
        msg_bit.extend([int(b) for b in binary_bit])
    return msg_bit

def bits2msg(msg_bits):
```

```
    msg = []
    for i in range(0, len(msg_bits), 8):
        byte = "".join(map(str, msg_bits[i:i + 8]))
        msg.append(chr(int(byte, 2)))
    return "".join(msg)


def hideMessage(filename, secret_message):
    cover_image = Image.open(filename).convert("RGB")
    width, height = cover_image.size
    message_bits = generateBits(secret_message)
    bit_index = 0
    #embed bits horizontal
    for y in range(height):
        for x in range(width):
            if bit_index < 100:
                r, g, b = cover_image.getpixel((x, y))
                # if color detected then embed messaage horizontal
                r = (r & ~1) | message_bits[bit_index]
                cover_image.putpixel((x, y), (r, g, b))
                bit_index += 1
            else:
                break
        if bit_index >= len(message_bits):
```

```
            break
    #embed bits vertical
    for x in range(width):
        for y in range(height):
            if bit_index < len(message_bits):
                r, g, b = cover_image.getpixel((x, y))
                # if B is two coloured then embed message
                g = (g & ~1) | message_bits[bit_index]
                cover_image.putpixel((x, y), (r, g, b))
                bit_index += 1
            else:
                break
        if bit_index >= len(message_bits):
            break
    return cover_image


def extractMessage(filepath):
    cover_image = Image.open(filepath)
    width, height = cover_image.size
    extracted_bits = []
    potential_message = ""
    for y in range(height):
        for x in range(width):
```

```
            #inspect colour in each pixel
            r, g, b = cover_image.getpixel((x, y))
            #if two colour then extract bit
            extracted_bits.append(r & 1)
            if len(extracted_bits) < 100:
                potential_message = bits2msg(extracted_bits[:len(extracted_bits)//8*8])  # Only consider full bytes
            else:
                break
        print(len(extracted_bits))
        if len(extracted_bits) >= 100:
            potential_message = potential_message.split("#")[0]
            break
    for x in range(width):
        for y in range(height):
            r, g, b = cover_image.getpixel((x, y))
            extracted_bits.append(g & 1)
            if len(extracted_bits) > 1000:
                potential_message = bits2msg(extracted_bits[:len(extracted_bits)//8*8])
                if "#" in potential_message:
                    potential_message = potential_message.split("#")[0]
                    break
        if "#" in potential_message:
            potential_message = potential_message.split("#")[0]
```

```
            break
    return potential_message


def ImageStegAction(request):
    if request.method == 'POST':
        global username
        message = request.POST.get('t1', False)
        message += " #"
        filename = request.FILES['t2'].name
        image_data = request.FILES['t2'].read()
        if os.path.exists("SecurityApp/static/files/"+filename):
            os.remove("SecurityApp/static/files/"+filename)
        with open("SecurityApp/static/files/"+filename, "wb") as file:
            file.write(image_data)
        file.close()
        dd = str(date.today())
        cover_image = hideMessage("SecurityApp/static/files/"+filename, message)
        os.remove("SecurityApp/static/files/"+filename)
        cover_image.save("SecurityApp/static/files/"+filename)
        with open("SecurityApp/static/files/"+filename, "rb") as file:
            encrypted_data = file.read()
        file.close()
        hashcode = hashlib.sha256(encrypted_data).hexdigest()
```

```
    db_connection = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root', password = 'root', database = 'cybersecurity',charset='utf8')

    db_cursor = db_connection.cursor()

    student_sql_query = "INSERT INTO files VALUES('"+username+"','"+filename+"','"+hashcode+"','"+dd+"','Steganography')"

    db_cursor.execute(student_sql_query)

    db_connection.commit()

    context= {'data':'<font size="3" color="blue">Message successfully hidden in given image</font>'}

    return render(request, 'ImageSteg.html', context)


def Download(request):

  if request.method == 'GET':

    global fileList

    name = request.GET.get('requester', False)

    with open("SecurityApp/static/files/"+name, "rb") as file:

      data = file.read()

    file.close()

    private_key, public_key = getECCKeys()

    decrypted_data = ECCDecrypt(data, private_key)

    private_key = generateKeys()

    decrypted_data = decryptAES(decrypted_data, private_key)

    response = HttpResponse(decrypted_data,content_type='application/force-download')

    response['Content-Disposition'] = 'attachment; filename='+name
```

```
    return response


def AccessData(request):
  if request.method == 'GET':
    global username
    output = '<table border=1 align=center width=100%><tr><th><font size="3" color="black">File Owner Name</th><th><font size="3" color="black">Filename</th>'
    output+='<th><font size="3" color="black">Hashcode</th><th><font size="3" color="black">Upload Date</th><th><font size="3" color="black">File Security Type</th>'
    output+='<th><font size="3" color="black">Steg Image</th><th><font size="3" color="black">Download File</th></tr>'
    con = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root', password = 'root', database = 'cybersecurity',charset='utf8')
    with con:
      cur = con.cursor()
      cur.execute("select * FROM files where username='"+username+"'")
      rows = cur.fetchall()
      for row in rows:
        name = row[0]
        fname = row[1]
        hashcode = row[2]
        upload_date = row[3]
        upload_type = row[4]
        output += '<tr><td><font size="3" color="black">'+str(name)+'</td><td><font size="3" color="black">'+str(fname)+'</td>'
```

```
            output+='<td><font size="3" color="black">'+hashcode+'</td>'
            output+='<td><font size="3" color="black">'+upload_date+'</td>'
            output+='<td><font size="3" color="black">'+upload_type+'</td>'
            if upload_type == "Steganography":
                output+='<td><img src="static/files/'+fname+'" height="200" width="200"/></td>'
                message = extractMessage("SecurityApp/static/files/"+fname)
                output+='<td><font size="3" color="blue">Hidden Message = '+message+'</td>'
            else:
                output+='<td><font size="3" color="red">--</td>'
                output +='<td><a href=\'Download?requester='+fname+'\'><font size=3 color=red>Download</font></a></td></tr>'
        output += "</table><br/><br/><br/><br/>"
        context= {'data':output}
        return render(request, 'UserScreen.html', context)


def ImageSteg(request):
    if request.method == 'GET':
        return render(request, 'ImageSteg.html', {})


def HybridEncryptionAction(request):
    if request.method == 'POST':
        global username
        myfile = request.FILES['t1'].read()
        fname = request.FILES['t1'].name
```

```
    dd = str(date.today())

    #get IBE key for file encryption

    private_key = generateKeys()

    encrypted_data = encryptAES(myfile, private_key)

    private_key, public_key = getECCKeys()

    encrypted_data = ECCEncrypt(encrypted_data, public_key)

    with open("SecurityApp/static/files/"+fname, "wb") as file:

        file.write(encrypted_data)

    file.close()

    hashcode = hashlib.sha256(encrypted_data).hexdigest()

    db_connection = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root', password =
'root', database = 'cybersecurity',charset='utf8')

    db_cursor = db_connection.cursor()

    student_sql_query = "INSERT INTO files
VALUES('"+username+"','"+fname+"','"+hashcode+"','"+dd+"','Hybrid Encryption')"

    db_cursor.execute(student_sql_query)

    db_connection.commit()

    context= {'data':'<font size="3" color="blue">Hybrid Encrypted file successfully saved at
server network</font>'}

    return render(request, 'HybridEncryption.html', context)


def HybridEncryption(request):

    if request.method == 'GET':

        return render(request, 'HybridEncryption.html', {})
```

```
def UserLogin(request):
    if request.method == 'GET':
        return render(request, 'UserLogin.html', {})


def index(request):
    if request.method == 'GET':
        return render(request, 'index.html', {})


def Register(request):
    if request.method == 'GET':
       return render(request, 'Register.html', {})


def RegisterAction(request):
    if request.method == 'POST':
        username = request.POST.get('t1', False)
        password = request.POST.get('t2', False)
        contact = request.POST.get('t3', False)
        email = request.POST.get('t4', False)
        address = request.POST.get('t5', False)
        status = "none"
        con = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root', password = 'root', database
= 'cybersecurity',charset='utf8')
        with con:
```

```
        cur = con.cursor()
        cur.execute("select username FROM register")
        rows = cur.fetchall()
        for row in rows:
            if row[0] == username:
                status = '<font size="3" color="blue">Username already exists</font>'
                break
    if status == "none":
        db_connection = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root', password =
'root', database = 'cybersecurity',charset='utf8')
        db_cursor = db_connection.cursor()
        student_sql_query = "INSERT INTO register
VALUES('"+username+"','"+password+"','"+contact+"','"+email+"','"+address+"')"
        db_cursor.execute(student_sql_query)
        db_connection.commit()
        print(db_cursor.rowcount, "Record Inserted")
        if db_cursor.rowcount == 1:
            status = '<font size="3" color="blue">Signup process completed</font>'
    context= {'data': status}
    return render(request, 'Register.html', context)


def sendOTP(email, otp_value):
    em = []
    em.append(email)
```

```
    with smtplib.SMTP_SSL('smtp.gmail.com', 465) as connection:
        email_address = 'kaleem202120@gmail.com'
        email_password = 'xyljzncebdxcubjq'
        connection.login(email_address, email_password)
        connection.sendmail(from_addr="kaleem202120@gmail.com", to_addrs=em, msg="Subject
: Your OTP : "+otp_value)


def UserLoginAction(request):
    if request.method == 'POST':
        global username, otp, email
        uname = request.POST.get('username', False)
        password = request.POST.get('password', False)
        index = 0
        con = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root', password = 'root', database
= 'cybersecurity',charset='utf8')
        with con:
            cur = con.cursor()
            cur.execute("select username, password, email FROM register")
            rows = cur.fetchall()
            for row in rows:
                if row[0] == uname and password == row[1]:
                    email = row[2]
                    username = uname
                    index = 1
```

```
            break
        if index == 1:
            otp = str(random.randint(1000, 9999))
            sendOTP(email, otp)
            context= {'data':'<font size="3" color="blue">OTP sent to your mail</font>'}
            return render(request, 'OTP.html', context)
        else:
            context= {'data':'<font size="3" color="blue">login failed</font>'}
            return render(request, 'UserLogin.html', context)


def OTPAction(request):
    if request.method == 'POST':
        global username, otp
        user_otp = request.POST.get('t1', False)
        if otp == user_otp:
            context= {'data':'<font size="3" color="blue">OTP Succesfully Validated<br/>welcome
'+username+'</font>'}
            return render(request, 'UserScreen.html', context)
        else:
            context= {'data':'<font size="3" color="blue">login failed</font>'}
            return render(request, 'OTP.html', context)
```

5. RESULTS & DISCUSSION

5. RESULTS AND DISCUSSION

The following screenshots showcase the results of our project, highlighting key features and functionalities. These visual representations provide a clear overview of how the system performs under various conditions, demonstrating its effectiveness and user interface. The screenshots serve as a visual aid to support the project's technical and operational achievements.

5.1 HOME PAGE

The home page of the Hybrid AES–ECC and Steganographic Data Security Framework, which serves as the primary entry point to the system. It provides access to essential navigation options such as user login and new user registration. The home page establishes the system scope before any secure operations are performed. It acts as the initial interface for authenticated and controlled user interaction.

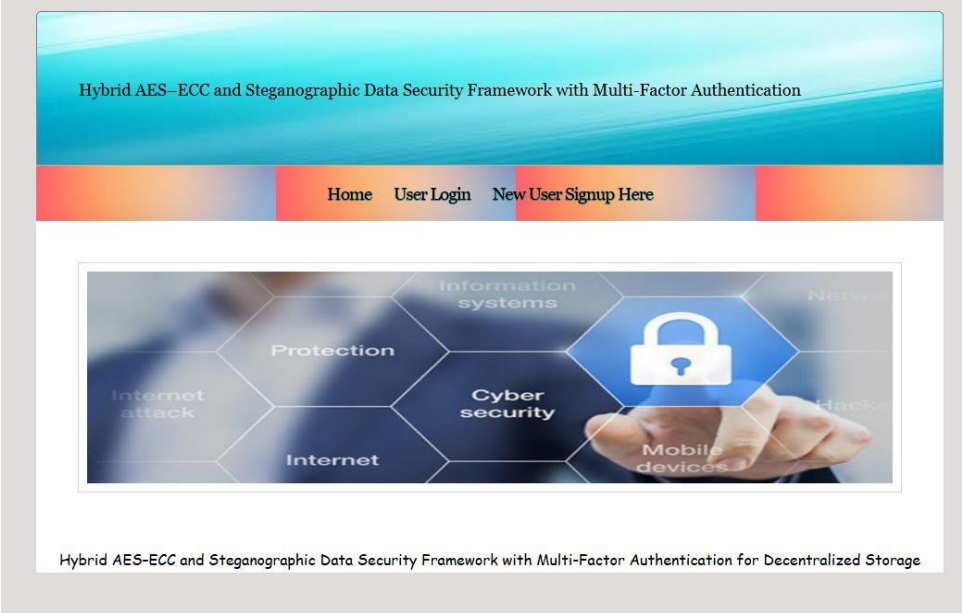

FIG 5.1: Home page

5.2 REGISTRATION PAGE

The user registration page designed for enrolling new users into the secure data protection system. This page allows users to submit personal and contact details required for authentication. The information collected during registration is stored securely in the database for future

verification. User registration ensures that only authorized individuals can access system functionalities. This step initiates the secure access workflow of the framework.

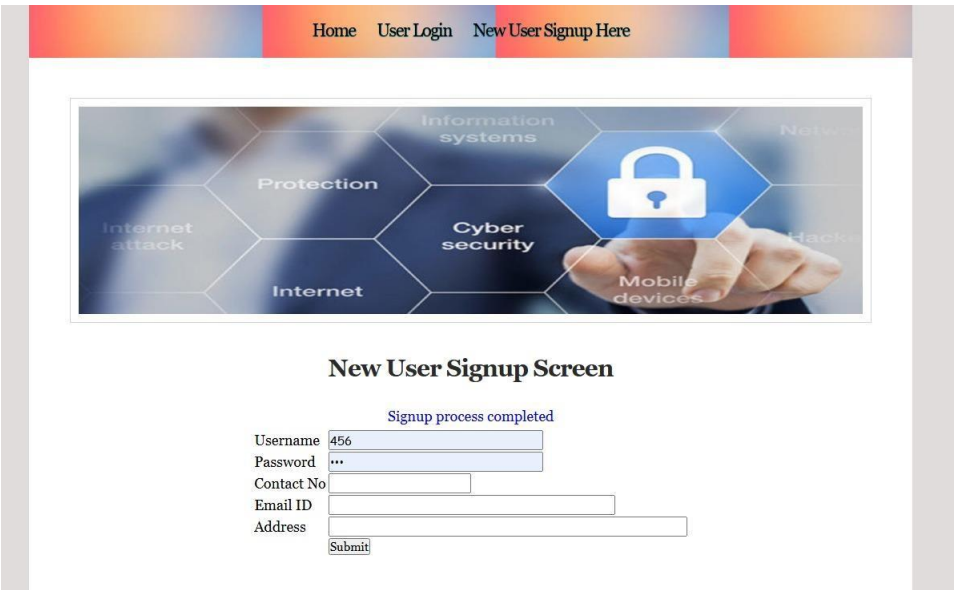

FIG 5.2: Registration page

5.3 OTP VERIFICATION PAGE

The OTP verification page that implements multi-factor authentication within the system. After successful credential validation, a one-time password is generated and sent to the registered email address. The user must enter the correct OTP to complete the authentication process. This mechanism strengthens security by preventing unauthorized logins.

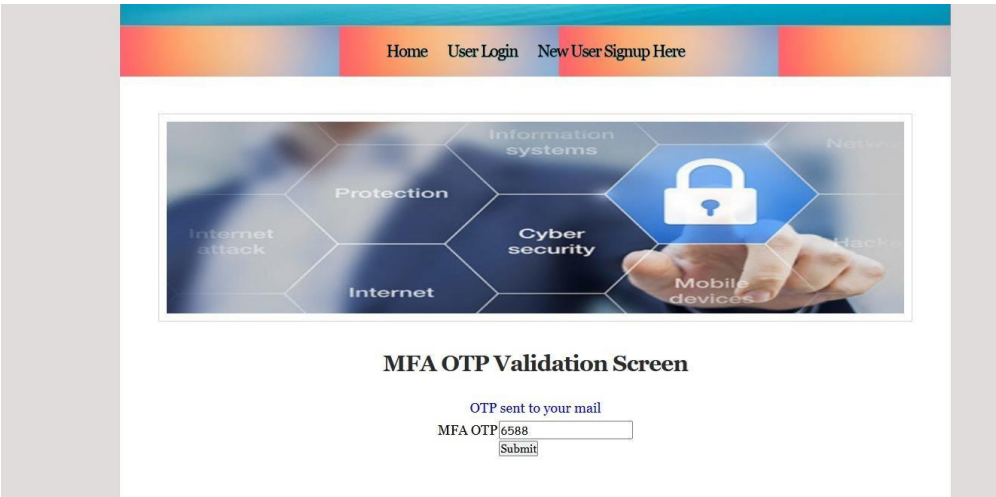

FIG 5.3: OTP Verification

5.4 LOGIN PAGE

The user login page of the data protection system following successful OTP validation. This page confirms authenticated user access and allows entry into the secured environment. It serves as the gateway to encryption, steganography, and data management modules. User sessions are established only after proper authentication. This ensures controlled and secure access to sensitive system resources.

FIG 5.4: Login Page

5.5 UPLOADING A FLE

The file upload interface for hybrid encryption-based data protection. Users upload data files that are encrypted using AES for confidentiality and ECC for secure key handling. The encrypted files are stored securely on the server along with integrity verification information.

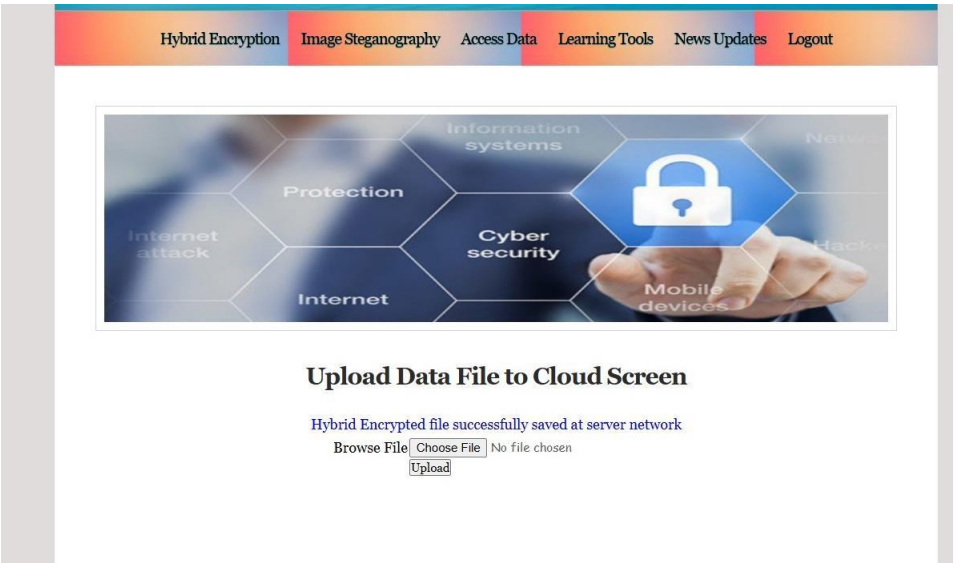

FIG 5.5: File Uploading

5.6 IMAGE STEGANOGRAPHY INTERFACE

The image steganography interface used for hidden data communication. Users input a secret message and select a digital image as the carrier.

FIG 5.6: Image Steganography

5.7 ACCESSING DATA

The data access page of the Hybrid Encryption and Steganography framework. It displays file metadata for verification purposes. Authorized users can download hybrid-encrypted files or view extracted hidden messages from stego-images.

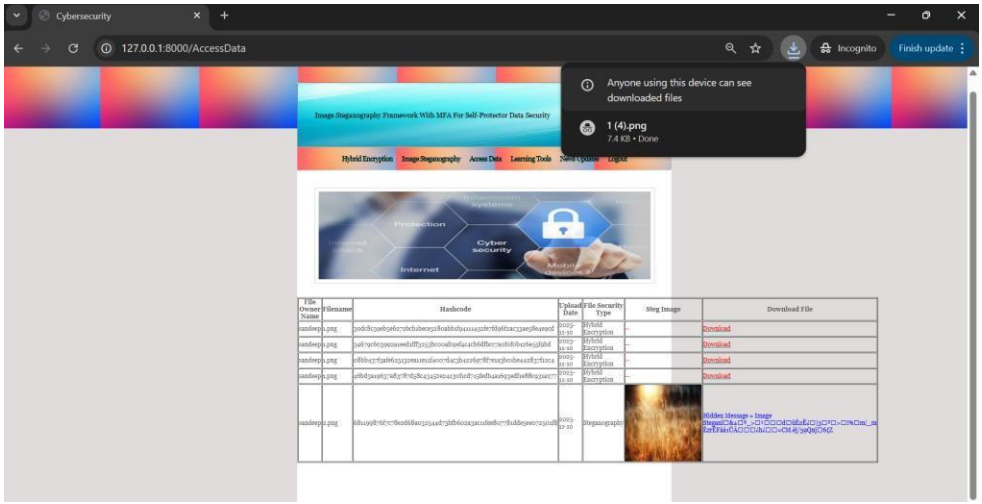

FIG 5.7: Accessing Data

5.8 LEARNING TOOL

The learning tool module integrated into the system for cybersecurity awareness and skill development. This module provides users with access to external cybersecurity learning platforms and resources.

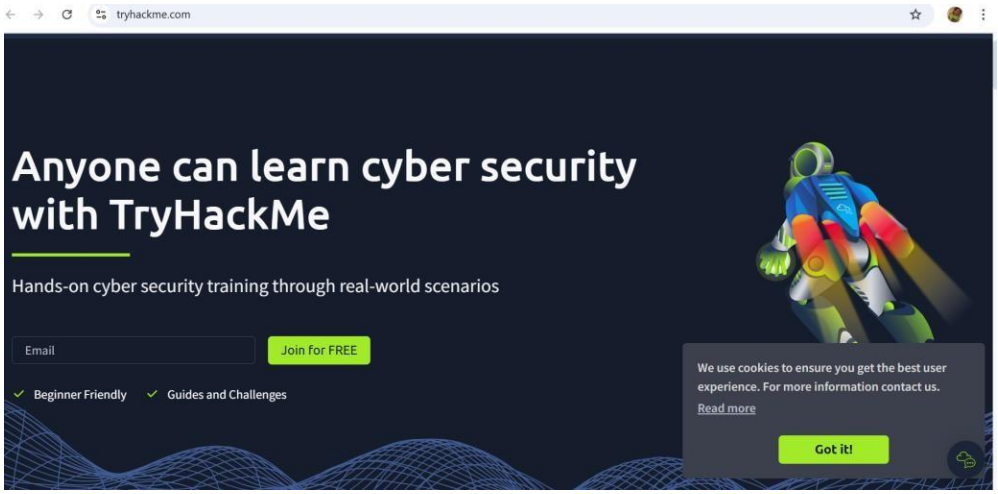

FIG 5.8: Learning Tool

5.9 NEWS UPDATE

The news update module designed to provide users with the latest cybersecurity-related information. This section presents real-time updates on data breaches, cyberattacks, vulnerabilities, and security trends. It helps users stay informed about emerging threats and industry developments. The module supports informed decision-making and security awareness.

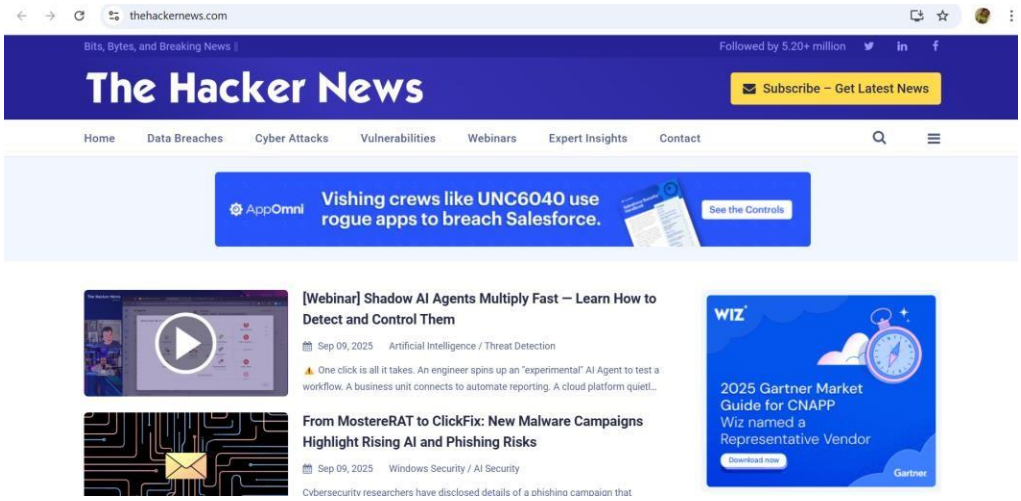

FIG 5.9: News Update

6. VALIDATION

6. VALIDATION

The validation of this project primarily relies on extensive testing and well-defined test cases to ensure the accuracy and effectiveness of the inappropriate content detection system. The testing process involves multiple stages, including dataset validation, model performance evaluation, and real-world testing. By implementing a structured validation approach, we can ensure that the system consistently delivers high accuracy in detecting inappropriate content while minimizing false positives and false negatives.

6.1 INTRODUCTION

In this hybrid data security framework, different testing procedures are carried out to ensure that the overall system operates effectively and achieves its objective of protecting sensitive information. The main focus is to verify that each stage of the framework, including authentication, encryption, steganographic embedding, data extraction, and decryption, performs accurately and efficiently. By examining these components, the reliability and effectiveness of the overall security mechanism can be ensured.

Initially, the input data is encrypted using the Advanced Encryption Standard (AES) algorithm, which converts the original data into secure ciphertext. The encrypted data is then embedded into a cover image using image steganography techniques. This process ensures that the confidential information is hidden within the image without causing noticeable changes to its visual quality. After embedding, the hidden data is extracted from the stego image and decrypted to confirm that the original information can be retrieved correctly without any loss of data.

Additionally, the authentication mechanism is examined to ensure that only authorized users are allowed to access and retrieve the secured information. Multi-Factor Authentication (MFA) strengthens the access control process by requiring valid login credentials and additional verification. The results obtained from these testing procedures demonstrate that the hybrid approach effectively enhances confidentiality, integrity, and secure data management within the framework.

6.2 TEST CASES

6.2.1 USER REGISTRATION

Test case ID	Test case name	Purpose	Input	Output
1	User Registration	To allow new users to create an account	User enters username, email and password	Account successfully created

6.2.2 USER LOGIN

Test case ID	Test case name	Purpose	Input	Output
1	Login Authentication	To verify user login credentials	Correct username and password entered	Login successful
2	Invalid Login	To check system response for incorrect login	Incorrect username or password	Error message displayed

6.2.3 AES ENCRYPTION

Test case ID	Test case name	Purpose	Input	Output
1	File Encryption	To convert plaintext data into ciphertext	User uploads file for encryption	Encrypted data generated successfully

6.2.4 IMAGE STEGANOGRAPHY EMBEDDING

Test case ID	Test case name	Purpose	Input	Output
1	Data Embedding	To hide encrypted data inside an image	Encrypted data and cover image	Stego image generated successfully

6.2.5 DATA EXTRACTION

Test case ID	Test case name	Purpose	Input	Output
1	Extract Hidden Data	To retrieve hidden encrypted data from image	Stego image selected	Encrypted data extracted successfully

6.2.6 DATA DECRYPTION

Test case ID	Test case name	Purpose	Input	Output
1	Data Decryption	To recover original data from ciphertext	Extracted encrypted data	Original data restored successfully

7. CONCLUSION & FUTURE ASPECTS

7. CONCLUSION AND FUTURE ASPECTS

In conclusion, the proposed framework successfully demonstrates an effective approach for securing sensitive information by integrating multiple security mechanisms. The implementation combines encryption and steganography techniques to provide enhanced protection against unauthorized access and data breaches. Through the use of Advanced Encryption Standard (AES), Elliptic Curve Cryptography (ECC), and image steganography, the framework ensures that confidential data remains protected during storage and transmission.

The overall results indicate that the hybrid approach provides a robust and efficient solution for modern data protection requirements. By combining cryptographic techniques with data hiding methods, the framework significantly enhances security while maintaining system performance and usability.

7.1 Project Conclusion

This work presents a hybrid data protection framework that integrates **AES encryption**, **ECC-based key protection**, **image steganography**, and **Multi-Factor Authentication (MFA)** to provide a secure environment for storing and transmitting confidential information. AES is utilized to encrypt the original data into secure ciphertext, while ECC is applied to protect the encryption keys and ensure secure key management. The encrypted information is then embedded into digital images using steganography techniques, enabling secure and concealed data transmission.

The implementation demonstrates that the framework effectively hides encrypted data within images without significantly altering the image quality. During retrieval, the hidden data can be successfully extracted and decrypted to restore the original information accurately. The authentication mechanism further ensures that only authorized users can access the protected data.

Overall, the hybrid framework enhances the level of data security by combining strong cryptographic algorithms with steganographic techniques. The results obtained from implementation and testing confirm that the proposed approach provides an efficient and reliable solution for secure data storage and communication in modern digital systems.

7.2 Future Scope

The proposed framework can be further improved by incorporating advanced technologies that enhance security, scalability, and usability. One important enhancement is the integration of **cloud-based storage**, which enables secure and scalable data management. By combining cloud services with hybrid encryption and steganography techniques, sensitive information can be stored, managed, and accessed efficiently while maintaining strong data protection.

Another promising enhancement is the integration of **blockchain technology** to improve transparency and data integrity. Blockchain can provide immutable records and secure audit trails for data access and transactions, ensuring that any unauthorized modification can be easily detected. This approach can significantly strengthen trust and reliability in secure data management.

The framework can also be expanded to support **mobile platforms and Internet of Things (IoT) devices**, enabling secure communication and data sharing in real-time environments. Such compatibility would allow the framework to be used in modern distributed systems where secure and efficient data exchange is essential.

Furthermore, the adoption of **advanced steganography techniques** and optimized encryption methods can improve the overall efficiency and robustness of the framework. These improvements may increase the data embedding capacity, enhance resistance to detection, and reduce processing time during encryption and decryption. With these developments, the framework can evolve into a more powerful and adaptable solution for secure data storage and communication.

8. BIBLIOGRAPHY

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8.2 GitHub Link

https://github.com/hajira1221/Major-project-Batch-6